Rohin Joshi

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Security Engineer Internship - Secure Distributed Systems, Red Team(Summer 2026)

US Citizen

EDUCATION

Carnegie Mellon University — Pittsburgh, PA, USA

Master of Science in Information Security (GPA - 3.8) — August 2025 - May 2027

- **INI TECH Fellow 2025 (**Entrepreneur fellowship)
- **Coursework**: Distributed Systems, Computer Networks, Information Security, Offensive Security, Computer Systems, Database Systems

SKILLS

- **Languages:** C, C++, Go, Python, Java, C#, JavaScript, Assembly
- **Security:** Vulnerability Research, Binary Exploitation, Cryptography, Secure Protocols, mTLS
- **Systems:** Distributed Systems, Concurrency, Operating Systems, Networking
- **Infra & Tools:** Docker, Git, Kafka, ZooKeeper, NATS, PostgreSQL, Redis, AWS, Azure, Wireshark, Ghidra, Burp Suite

SECURITY EXPERIENCE

- **CVE-2025-11201(1-Day):** Achieved Remote Code Execution in MLflow Tracking Server via directory traversal, unsafe pickle deserialization, and Python site customization, enabling arbitrary code execution on the host.
- **CVE-2025-10283 / 10284(1-Day):** Achieved RCE in Bbot by exploiting unsanitized Git index handling (gitDumper) and unsafe archive extraction in the unarchive module, including 0-click RCE via crafted RAR files.
- **Binary Exploitation:** Exploited heap overflows using heap spraying,and double free mechanism, and buffer overflows via ret2libc and ROP gadgets; bypassed ASLR, NX, and stack canaries through memory layout manipulation and allocator behavior analysis.
- **Cryptography & Web Security:** Analyzed and exploited weaknesses in AES-CTR, AES-CBC, ECB, SHA-256 length extension, OTP misuse, SQL injection, XSS, CSRF, and certificate chain validation failures.
- **Blockchain Lead at Metastart (**Startup,2022-2023): Led the cryptography and blockchain research wing at Metastart focussing on cryptography mechanisms such as zero knowledge proofs, merkle trees and self sovereign identities.
- **Competitive & OSS:** Top 5 in CMU Hacking & Offensive Security CTF; contributor to BCoin (BIP-157/158 – Golomb-Rice coding); 2× blockchain hackathon winner.

EXPERIENCE

Borg Markkula Oy — Helsinki, Finland

SOFTWARE ENGINEER — September 2024 - July 2025

- Designed and implemented a parallel execution engine for large-scale data processing using OpenMP task + loop parallelism and a thread-pool scheduling model, reducing end-to-end latency from 650 ms → 30 ms under high concurrency.
- Optimized shared-memory concurrency and cache coherence via cache-aware data layouts, memory batching, and reduced cross-core contention, enabling predictable performance on multi-core systems.

Indian Institute of Science — Bangalore, India

RESEARCH ASSISTANT — January 2023 - December 2024

- Led a project in collaboration with TCS Research to design KAGS, a network-of-networks architecture for hierarchical reasoning with LLM agents, introducing validation and specialization layers to reduce hallucinations and untrusted output propagation.reduced design generation time by 80% through system-level architectural optimization.
- Published KAGS in Advanced Engineering Informatics (Q1 journal with impact factor 9.9)

Unisys, Office of the CTO — Bangalore, India

SOFTWARE ENGINEERING INTERN — June 2022 - December 2022

- Designed and deployed mTLS authentication across distributed DBoM nodes, enforcing certificate-based identity, encrypted transport, and mutual trust across services.
- Built a secure, scalable pub-sub WebSocket system supporting 200K+ concurrent clients while reducing notification latency by 75%.
- Implemented an Azure-based async ingestion and indexing pipeline for financial filings, focusing on secure ingestion, isolation, and fault-tolerant processing for internal RAG tools.

PROJECTS

- **Bustub Database(C++, Database systems):** Developing a disk-oriented DBMS in C++ from scratch, implementing an ARC-based buffer pool manager, thread-safe B+Tree indexing, and a query execution engine with rule-based optimization and MVOCC transaction support. Held top 5 positions in CMU DB leaderboard.
- **Hierarchical Key-Value Cluster(Go):** Designed and implemented a consistent, sharded hierarchical key-value store with list/create/delete operations, integrating Raft for replicated logs and leader election per directory shard and testing under simulated network faults.